

D1 Mensuration — Worksheet

IGCSE Mathematics 0580 | Practice Problems

D1_Q1: Perimeter of 2D Shapes — Problems

F

Foundation

[1 mark]

Q1. Find the perimeter of a rectangle with length 8 cm and width 5 cm.

F

Foundation

[1 mark]

Q2. Find the perimeter of a square with side length 6 cm.

F

Foundation

[1 mark]

Q3. A triangle has sides of lengths 7 cm, 5 cm, and 9 cm. Find its perimeter.

F

Foundation

[1 mark]

Q4. Find the circumference of a circle with radius 7 cm (use $\pi = \frac{22}{7}$).

C

Core

[1 mark]

Q5. A rectangle has perimeter 34 cm and length 12 cm. Find its width.

C Core [1 mark]

Q6. A regular hexagon has perimeter 48 cm. Find the length of one side.

C Core [1 mark]

Q7. Find the circumference of a circle with diameter 14 cm (use $\pi = 22/7$).

C Core [1 mark]

Q8. Find the perimeter of a semicircle with radius 7 cm including the diameter edge (use $\pi = 22/7$).

C Core [1 mark]

Q9. An L-shape has sides: 10 m, 6 m, 4 m, 3 m, 6 m, and 3 m. Find its perimeter.

C Core [1 mark]

Q10. A circular wheel has radius 35 cm. How far does it travel in 100 complete rotations? (use $\pi = 22/7$, answer in metres)

C Core [1 mark]

Q11. A triangle has perimeter 30 cm. Two sides are 11 cm and 8 cm. Find the third side.

C

Core

[1 mark]

Q12. A regular octagon has side length 4.5 cm. Find its perimeter.

H

Challenge

[1 mark]

Q13. A rectangle has length $(3x + 2)$ cm and width $(x + 1)$ cm. Find an expression for its perimeter.

H

Challenge

[1 mark]

Q14. A shape consists of a rectangle 12 cm by 8 cm with a semicircle (radius 4 cm) on one of the longer sides. Find the total perimeter (use $\pi = 3.142$).

H

Challenge

[1 mark]

Q15. A sector of a circle has radius 10 cm and arc length 15.71 cm. Find the angle of the sector in degrees (use $\pi = 3.142$).

H

Challenge

[1 mark]

Q16. A shape consists of a square of side $(x+3)$ cm and an equilateral triangle of side x cm sharing a side. Find the total perimeter in terms of x .

D1_Q2: Area of 2D Shapes — Problems

F

Foundation

[1 mark]

Q17. Find the area of a rectangle with length 9 cm and width 7 cm.

F

Foundation

[1 mark]

Q18. Find the area of a square with side length 8 cm.

F

Foundation

[1 mark]

Q19. Find the area of a triangle with base 12 cm and perpendicular height 5 cm.

F

Foundation

[1 mark]

Q20. Find the area of a circle with radius 7 cm (use $\pi = 22/7$).

C

Core

[1 mark]

Q21. Find the area of a parallelogram with base 10 cm and perpendicular height 6 cm.

C

Core

[1 mark]

Q22. A trapezium has parallel sides of length 8 cm and 12 cm, and perpendicular height 5 cm. Find its area.

C Core [1 mark]

Q23. Find the area of a circle with diameter 14 cm (use $\pi = 22/7$).

C Core [1 mark]

Q24. Find the area of a semicircle with radius 10 cm (use $\pi = 3.142$).

C Core [1 mark]

Q25. A triangle has area 42 cm^2 and height 7 cm. Find its base.

C Core [1 mark]

Q26. Convert 5 m^2 to cm^2 .

C Core [1 mark]

Q27. A rectangle has area 96 cm^2 and width 8 cm. Find its length.

C Core [1 mark]

Q28. A triangular flag has base 40 cm and height 25 cm. Find its area.

H

Challenge

[1 mark]

Q29. A trapezium has area 60 cm^2 , height 6 cm, and one parallel side 8 cm. Find the other parallel side.

H

Challenge

[1 mark]

Q30. A rectangle has length 15 cm and width 10 cm. The length is increased by 20% and the width is decreased by 20%. Find the percentage change in area.

H

Challenge

[1 mark]

Q31. A sector has radius 8 cm and angle 60 degrees. Find its area (use $\pi = 3.142$).

H

Challenge

[1 mark]

Q32. A trapezium has parallel sides $(x+3)$ and $(2x+1)$ cm, and height 4 cm. Write an expression for its area.

D1_Q3: Composite Shapes — Problems

F

Foundation

[1 mark]

Q33. An L-shaped figure consists of a 5 m by 4 m rectangle joined to a 3 m by 2 m rectangle. Find the total area.

F

Foundation

[1 mark]

Q34. A shape consists of a rectangle 6 m by 4 m with a triangle of base 6 m and height 3 m on top. Find the total area.

F

Foundation

[1 mark]

Q35. Find the perimeter of the L-shape with outer sides: 7 m, 4 m, 3 m, 2 m, 4 m, 2 m.

F

Foundation

[1 mark]

Q36. A rectangular lawn 10 m by 8 m has a rectangular flowerbed 3 m by 2 m cut out from it. Find the remaining area.

C

Core

[1 mark]

Q37. Find the area of a ring with outer radius 8 cm and inner radius 5 cm (use $\pi = 3.142$).

C

Core

[1 mark]

Q38. A rectangle 10 m by 6 m has a semicircle of radius 3 m attached to one short side. Find the total area (use $\pi = 3.142$).

C Core

[1 mark]

Q39. An L-shape has total width 12 m, and two vertical segments of 7 m and 5 m. The horizontal segments are 5 m and the remaining part. Find the missing horizontal segment.

C Core

[1 mark]

Q40. A rectangular garden 15 m by 10 m has a path of width 1 m around the inside. Find the area of the path.

C Core

[1 mark]

Q41. A figure is made of a rectangle 8 cm by 6 cm with a right-angled triangle of base 8 cm and height 6 cm on top. Find the total area.

C Core

[1 mark]

Q42. A shape has a rectangle 12 cm by 5 cm with a semicircle (radius 2.5 cm) on one of the 5 cm sides. Find the total perimeter (use $\pi = 3.142$).

C Core

[1 mark]

Q43. A shape consists of a rectangle 10 m by 4 m with a right-angled triangle of base 3 m and height 4 m attached to one short side. Find total area.

C

Core

[1 mark]

Q44. A composite L-shape has sides: 15 m, 8 m, 7 m, 3 m, 8 m, and one unknown side. Find the unknown side if the perimeter is 46 m.

H

Challenge

[1 mark]

Q45. A shape consists of a rectangle of width x m and height 4 m with a semicircle of radius $x/2$ m on one of the longer sides. Write an expression for the total area in terms of x .

H

Challenge

[1 mark]

Q46. A square of side 10 cm has four quarter-circles of radius 5 cm cut from its corners. Find the remaining area (use $\pi = 3.142$).

H

Challenge

[1 mark]

Q47. A circular pond of radius 5 m is surrounded by a path of width 1 m. Find the area of the path (use $\pi = 3.142$).

H

Challenge

[1 mark]

Q48. An L-shape has two rectangles: one is a m by b m, the other is c m by d m. Write an expression for the total area.

D1_Q4: Surface Area of Prisms — Problems

F

Foundation

[1 mark]

Q49. Find the surface area of a cuboid with length 5 cm, width 4 cm, and height 3 cm.

F

Foundation

[1 mark]

Q50. Find the surface area of a cube with side length 6 cm.

F

Foundation

[1 mark]

Q51. Find the surface area of a cylinder with radius 5 cm and height 8 cm (use $\pi = 3.142$).

F

Foundation

[1 mark]

Q52. A triangular prism has triangular ends with base 6 cm and height 4 cm, and length 10 cm. The rectangular faces are 6×10 , 5×10 , and 5×10 . Find the total surface area.

C

Core

[1 mark]

Q53. A cuboid has surface area 214 cm^2 , length 7 cm, and width 5 cm. Find its height.

C Core [1 mark]

Q54. Find the surface area of an open cylinder (no top) with radius 4 cm and height 10 cm (use $\pi = 3.142$).

C Core [1 mark]

Q55. A rectangular box measures 20 cm by 15 cm by 12 cm. How much cardboard is needed to make the box (ignoring overlaps)?

C Core [1 mark]

Q56. A cylindrical can has radius 3.5 cm and height 12 cm. Find the total surface area (use $\pi = 22/7$).

C Core [1 mark]

Q57. A prism has a cross-section which is a right-angled triangle with legs 3 cm and 4 cm. The prism length is 8 cm. Find the total surface area.

C Core [1 mark]

Q58. A cube has side 3 cm. If the side is doubled to 6 cm, by what factor does the surface area increase?

Core

[1 mark]

Q59. A cylinder has diameter 10 cm and height 15 cm. Find its surface area (use $\pi = 3.142$).

Core

[1 mark]

Q60. An L-shaped prism has a cross-section area of 22 cm^2 and perimeter of 24 cm . The prism length is 6 cm . Find its surface area.

Challenge

[1 mark]

Q61. A closed cylinder has volume 500 cm^3 and radius r . Write an expression for its surface area in terms of r .

Challenge

[1 mark]

Q62. A cuboid has dimensions $(x+2)$, (x) , and $(x-1)$ cm. Find an expression for its surface area in expanded form.

Challenge

[1 mark]

Q63. A solid consists of a cylinder of radius 4 cm and height 6 cm with a hemisphere of the same radius on top. Find the total surface area (use $\pi = 3.142$).

H	Challenge	[1 mark]
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Q64. A cuboid has sides x , $(x+1)$, and $(x+2)$ cm. Find an expression for its surface area.

D1_Q5: Volume of Prisms and Cylinders — Problems

F	Foundation	[1 mark]
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Q65. Find the volume of a cuboid with length 6 cm, width 5 cm, and height 4 cm.

F	Foundation	[1 mark]
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Q66. Find the volume of a cube with side length 7 cm.

F	Foundation	[1 mark]
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Q67. Find the volume of a cylinder with radius 4 cm and height 10 cm (use $\pi = 3.142$).

F	Foundation	[1 mark]
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Q68. A triangular prism has triangular cross-section with base 8 cm and height 5 cm, and length 12 cm. Find its volume.

C Core [1 mark]

Q69. A cuboid has volume 360 cm^3 , length 12 cm, and width 5 cm. Find its height.

C Core [1 mark]

Q70. A rectangular tank measures 50 cm by 30 cm by 40 cm. How many litres of water can it hold? (1 litre = 1000 cm^3)

C Core [1 mark]

Q71. A cylinder has diameter 14 cm and height 20 cm. Find its volume (use $\pi = 22/7$).

C Core [1 mark]

Q72. A prism has cross-sectional area 25 cm^2 and length 9 cm. Find its volume.

C Core [1 mark]

Q73. A cylindrical water tank has radius 30 cm and height 100 cm. How many litres does it hold? ($\pi = 3.142$, 1 L = 1000 cm^3)

C Core [1 mark]

Q74. A swimming pool is 25 m long, 10 m wide, and has an average depth of 1.5 m. Find its volume in m^3 .

C Core

[1 mark]

Q75. A cylinder has volume 1000 cm^3 and height 12 cm. Find its radius (use $\pi = 3.142$).

C Core

[1 mark]

Q76. Convert 2.5 m^3 to cm^3 .

H Challenge

[1 mark]

Q77. A cylinder of radius 6 cm and height 12 cm has a hemisphere of the same radius removed from the top. Find the remaining volume (use $\pi = 3.142$).

H Challenge

[1 mark]

Q78. A prism has cross-sectional area $(3x+2) \text{ cm}^2$ and length $(x+4) \text{ cm}$. Find an expression for its volume.

H Challenge

[1 mark]

Q79. A prism has a trapezium cross-section with parallel sides 5 cm and 9 cm, height 4 cm, and prism length 10 cm. Find its volume.

H

Challenge

[1 mark]

Q80. A cylindrical barrel of radius 30 cm contains water to a depth of 40 cm. How many litres of water are in the barrel? ($\pi = 3.142$, $1 \text{ L} = 1000 \text{ cm}^3$)

D1_Q6: Volume of Pyramids and Cones — Problems

F

Foundation

[1 mark]

Q81. Find the volume of a rectangular pyramid with base length 6 cm, width 5 cm, and height 9 cm.

F

Foundation

[1 mark]

Q82. Find the volume of a cone with radius 3 cm and height 7 cm (use $\pi = 22/7$).

F

Foundation

[1 mark]

Q83. Find the volume of a square-based pyramid with base side 4 cm and height 6 cm.

F

Foundation

[1 mark]

Q84. Find the volume of a cone with diameter 10 cm and height 12 cm (use $\pi = 3.142$).

C Core

[1 mark]

Q85. A rectangular prism and a pyramid have the same base (8 cm by 5 cm) and same height (12 cm). How many times larger is the prism's volume?

C Core

[1 mark]

Q86. A cone has volume 100 cm^3 and radius 5 cm. Find its height (use $\pi = 3.142$).

C Core

[1 mark]

Q87. A pyramid has a triangular base with base 6 cm and height 8 cm. The pyramid height is 10 cm. Find its volume.

C Core

[1 mark]

Q88. An ice cream cone has radius 3 cm and height 15 cm. Find its volume (use $\pi = 3.142$).

C Core

[1 mark]

Q89. A square pyramid has volume 200 cm^3 and height 12 cm. Find the side length of its base.

C Core

[1 mark]

Q90. A cone of radius 9 cm and height 18 cm is cut 6 cm from the top to form a frustum. The top cone (removed) has radius 3 cm and height 6 cm. Find the volume of the frustum (use $\pi = 3.142$).

C

Core

[1 mark]

Q91. A cone has radius 4 cm and height 9 cm. If both radius and height are doubled, by what factor does the volume increase?

C

Core

[1 mark]

Q92. A rectangular pyramid has base 10 cm by 6 cm and volume 180 cm^3 . Find its height.

H

Challenge

[1 mark]

Q93. A cone with radius 6 cm and a hemisphere of the same radius have equal volumes. Find the height of the cone (use $\pi = 3.142$).

H

Challenge

[1 mark]

Q94. A frustum of a cone has top radius 2 cm, bottom radius 5 cm, and height 9 cm. Find its volume (use $\pi = 3.142$).

H

Challenge

[1 mark]

Q95. Pyramid A has base 6 cm by 6 cm and height 10 cm. Pyramid B has base 12 cm by 12 cm and height 20 cm. What is the ratio of their volumes?

H

Challenge

[1 mark]

Q96. A cone of radius 5 cm and height 15 cm is full of water. Water is poured out until the depth is 10 cm. What fraction of the water remains?

D1_Q7: Volume of Spheres and Composite Solids — Problems

F

Foundation

[1 mark]

Q97. Find the volume of a sphere with radius 6 cm (use $\pi = 3.142$).

F

Foundation

[1 mark]

Q98. Find the volume of a hemisphere with radius 3 cm (use $\pi = 3.142$).

F

Foundation

[1 mark]

Q99. Find the surface area of a sphere with radius 5 cm (use $\pi = 3.142$).

F

Foundation

[1 mark]

Q100. Find the volume of a sphere with diameter 10 cm (use $\pi = 3.142$).

C Core

[1 mark]

Q101. A solid consists of a cylinder of radius 4 cm and height 10 cm with a hemisphere of the same radius on top. Find the total volume (use $\pi = 3.142$).

C Core

[1 mark]

Q102. A sphere has volume $288\pi \text{ cm}^3$. Find its radius.

C Core

[1 mark]

Q103. A cone of radius 3 cm and height 5 cm sits on top of a cylinder of the same radius and height 8 cm. Find the total volume (use $\pi = 3.142$).

C Core

[1 mark]

Q104. Sphere A has radius 3 cm and Sphere B has radius 9 cm. How many times greater is the volume of Sphere B?

C Core

[1 mark]

Q105. A capsule consists of a cylinder of radius 2 cm and length 8 cm with a hemisphere at each end. Find the total volume (use $\pi = 3.142$).

C

Core

[1 mark]

Q106. A sphere of radius 6 cm is melted and recast into a cylinder of radius 3 cm. Find the height of the cylinder (use $\pi = 3.142$).

C

Core

[1 mark]

Q107. Find the total surface area of a hemisphere (including the circular base) with radius 7 cm (use $\pi = 22/7$).

C

Core

[1 mark]

Q108. A spherical water tank has radius 1.5 m. How many litres of water can it hold? ($1 \text{ m}^3 = 1000 \text{ L}$, $\pi = 3.142$)

H

Challenge

[1 mark]

Q109. A cylinder and a cone have the same radius r and the same height h . A sphere has the same volume as the cylinder. Express the sphere's radius in terms of r and h .

H

Challenge

[1 mark]

Q110. A solid consists of a hemisphere of radius r on top of a cylinder of radius r and height r . Find the total volume in terms of r .

HChallenge[1 mark]

Q111. A sphere of radius 6 cm fits exactly inside a cylinder (touching the sides, top, and bottom). Find the volume of the space between them (use $\pi = 3.142$).

HChallenge[1 mark]

Q112. Three spheres of radius 2 cm each are placed in a cylinder of radius 2 cm. The spheres just fit one on top of another (height 12 cm). Find the volume of the empty space (use $\pi = 3.142$).